INTELLIGENT PREDICTION SOFTWARE FOR ROOM ACOUSTICS AND PSYCHOACOUSTICS USING NEURAL NETWORKS

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Comfort and wellbeing of an individual in a room depends largely on sound pressure level within the room and psychoacoustics. The importance of accurate prediction of acoustics parameters is vital during the initial room's design stage. Many researchers provide a detailed study on the interaction between the acoustics and psychoacoustics parameters. This paper develops a prediction model using different Neural Networks (NNs) such as Autoencoder, Support Vector Regression, Extreme Learning Machines, Deep Neural Network and others to predict the sound pressure level (SPL), Loudness and Sharpness. The results shown that the Deep Neural Network can predict the root mean square error of less than 2 dBA in SPL, 2 sone in Loudness and 3 acum in Sharpness.
Keywords: intelligent prediction, room acoustic software, psychoacoustics, neural network

1. Introduction

Machine Learning involves classification and regression. Most of the applications of machine learning in acoustics involved classification. The authors[1] had investigated the speech spectrum for the classification of audio for emotions and autism disorders. The paper examined the K-Nearest Neighbour (KNN), Support Vector Machine(SVM) models with the help of OpenSMILE feature extractor. In addition, the author uses Acoustics Segment Model (ASM) classification, incorporating temporal speech signal information for the classification using Deep Neural Network(DNN). The final model integrated the results of the models and made the classification accordingly. The system outperformed the baseline challenge for emotion and autism tasks. An overview of speech recognition[2] was discussed where the author compares Autoencoders, Convolutional Neural Network (CNN), Restricted Bolzmann Machine and Deep Belief Nets on phone's and continuous speech recognition. The results outperformed the conventional models of speech recognition using hidden Markov model(HMM). A systematic test for the DNNs for automatic speech recognition [3] was discussed where best practices for speech and language understanding tasks were examined. The systematic multi-step approach includes the selection of loss function, number of hidden layers and activation functions, optimization algorithms, dropout regularisation, and early stopping.

Application of neural network in the area of binaural room impulse response spatial analysis [4] was also examined. Data obtained from dummy binaural heads were used to test the cascade-correlation neural network. The results found that the predictions accuracy was about 65%. It was around 5% range of the expected values from KEMAR and KU100 for non-reflected sound. The accuracy of the model decreases due to low signal to noise ratios and misalignment in the measurements. Another paper [5] uses a neural network to predict the acoustics parameters such as SPL, Clarity and lateral energy of a large room. The author trained the model with 72 unoccupied concert halls data from various countries. The results were compared with the measurements. The results showed about 1 dBA difference in SPL, 0.5dB and 0.05 for Clarity and Lateral energy, respectively. Recently, the machine learning application [6] for the room acoustics parameter estimation was studied. The paper used the outputs from the geometrical models and wave-based models to estimate the reverberation time and early decay time as input to a three-layer neural network of a single room data.

Hence, the application of machine learning in acoustics is not new. It has been used in a wide variety of sound classification, modifications, and time-series prediction. Instead of using the simulated data, the results obtained from multiple rooms using the validated Class 1 equipment (such as SQoBold) was used to train the NN models. The data were collected from different rooms.

In summary, the paper is structured as follows. The details of the data collection are explained in section 2. The different machine learning models are explained briefly in section 3. The results are discussed in section 4 followed by conclusions.

2. Measurements and data collection

The data were obtained from different room types such as classrooms, office spaces, labs and accommodation spaces. In summary, the data (downloadable at https://github.com/mcschin1/room_acoustics) consists of room's parameters such as dimensions, absorption coefficients of each wall, noise source, distance from the noise source, temperature, air pressure, humidity, SPL, Loudness and Sharpness in different octave band frequency.

- Length (m)×Width (m)× Height (m) of room
- Absorption coefficient (front)
- Absorption coefficient (back)
- Absorption coefficient (right)
- Absorption coefficient (left)
- Absorption coefficient (ceiling)
- Absorption coefficient (floor)
- Number of air-con diffuser
- Octave frequency
- Noise level of dominating sound source (dBA)
- Distance from dominating sound source to measuring point
- Amount of furnishing/sound absorbing materials in room (1: Low, 2: Medium, 3: High)
- Atmospheric Temperature (°C)
- Air Pressure (hPa)
- Humidity (%)
- SPL
- Loudness (N5)
- Sharpness (S5)

There are a total of 18 input parameters and 3 outputs. A total of 2817 dataset for the machine learning model were tabulated in EXCEL sheet. The measurement guidelines and measurement procedures are explained below. The measurement was performed according to the ISO 3741. The following 15-point measurement array was used. The layout of the microphone placement is shown in Figure 1.

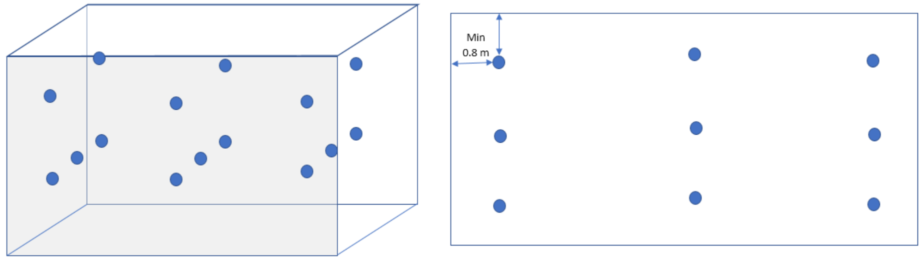


Figure 1: 3D view (left) and top view (right) of microphone positions layout

Sound Pressure level measurement was performed under specific environmental conditions (anechoic or reverberant chamber). For the quality and validity of the measurement results, a set of ISO standards governs the measurement was used. The ISO standards for noise measurements are classified into sub-categories according to the accuracy delivered. For example, the precision method gives most accurate and lowest uncertainty in the values[7] was used in this study. The ISO 3741 discusses the methodology of measuring the SPL for computing the sound energy and sound power from a reverberant room. The standard can be applied to all the types of noise including steady, unsteady, isolated bursts, fluctuating and etc. This standard gives the highest grade of accuracy in measurement, [8] by following the requirements.

- Source volume needs to be less than 2 % than of the reverberant test room.
- Reverberation room has to satisfy the standard specifications.
- Background noise level cannot exceed the defined values
- Instrumentation need to satisfy the requirements of the International Electrotechnical Commission (IEC) 61672-1:2002, Class 1
- Reference source (if applicable) must satisfy ISO 6926

Depending on the chosen measurement surface, the number of required microphone positions will vary. For the classroom, a 15-point measurement system was used as shown in Figure 1. As shown in Figure 1, the software developed by PyQt5 was used to train the measurement data. Figure 1 shows the setup for one such actual measurement. All data were pre-processed before used for the training in different algorithms. The input variables were normalized using the minimax function to a range of -1 to 1.

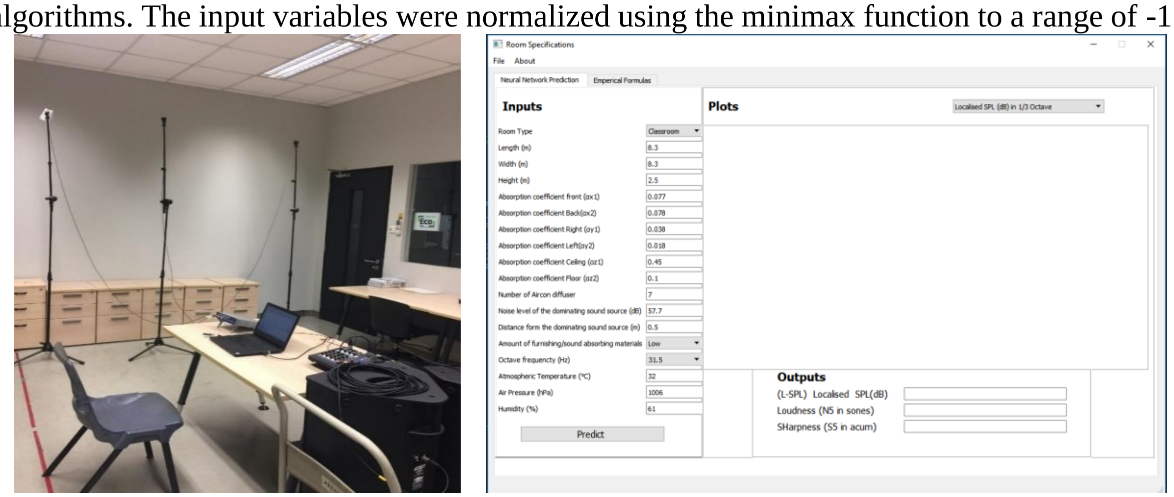


Figure 1: Measurement setup in researcher room and software used for machine learning

3. Brief descriptions of various neural networks

The data set was trained with different neural networks. The root mean square error (RMSE) was used to compare the different models. Around 70% of the data were used for training and remaining dataset were used for validating. The different NN models used are briefly explained.

1) Feedforward Neural Networks with classical backpropagation
2) Logistics Regression
3) K-Nearest Neighbours
4) Decision Tree
5) Support Vector Machine
6) Extreme Learning Machine
7) Random Forest
8) Auto Encoders
9) SoftMax
10) Deep Neural Network

Linear Regression model[9] is one of the probability-based model. It is a statistical model that returns the outcome based on the dataset with one or more independent variables. The goal of the logistics regression model is to determine the best fitting model that defines the relationship between the inputs (independent variables) and results (dependent variables). The model generates the coefficients of probabilities that characterize the significance and error.

K-Nearest Neighbors[10] uses a similarity measure (distance functions). The implementation requires to calculate the average weighted distance of K-nearest neighbors using the distance functions. A small value for K tends to overfit the data that generates a high error rate or vice versa. Hence, an optimal value of K has to be determined. Decision tree regression model [11] searches every distant value of its inputs(predictors) and split the values that return the minimum of Sum of Squared Error (SSE) statistics. The predictor values are recursively split within the groups named S1 and S2. The certain threshold is set for the model to stop splitting the sample size within the group. The over-fitting is prevented by penalizing the SSE by pruning the constructed tree over the tree size. Comparing the model with linear regression, the decision tree model calculates the importance of predictors rather than the coefficient of predators. The summation of the overall reduction of SSE (optimization criteria) defines the relative importance of predictors.

Random forest[12] is another model from the decision tree. The model splits the dataset into different small datasets and train different decision tree with each dataset and combines the results to make a forest. With more decision trees, the model becomes more robust. The main advantage of the random forest model is its flexibility to use in regression or classification problem. The model can deal with the issue of overfitting as the dataset is allocated in many different trees. As the model becomes more complex, the model is slow and thus not relevant for real-time applications. Support vector regression[13] is an extension to the Support vector machine for regression application. The architecture is similar to the Linear Regression model. It can be used to predict more complex data.

Autoencoder[14] was developed for unsupervised learning algorithms. It is mainly used for classification, dimensionality reduction, and noise reduction problems. Autoencoder aims to replicate the input to output. The hidden layer is compressed latent space representation of the input where it is then reconstructed in the output layer. The network composed of two mian layers namely, encoder and decoder. Encoder part of the network is responsible for compressing the input data to latent space representation while the decoder is responsible for the reconstruction for the output. SoftMax regression [15] model is a generalization of logistic regression that uses multi-class classification. The assumption is that the classes are mutually exclusive. The Logistics regression is a linear model mainly used in binary classification tasks. The model improves the performance by improving or minimizing the sigmoid function (logistic function) instead of minimizing the cost functions such as sum of square errors.

Extreme learning machines (ELM) reduces the training time of a network and improves the generalization performance [16]. Traditionally, the neural network models randomly assigned the training parameters such as weights, bias and use an iterative algorithm to improve and update these parameters. The ELM sets the weights and bias of the hidden neurons randomly; the output weights are estimated by the least-square method such as Moore-Penrose pseudo inverse. Deep neural network(DNN) [17] is necessarily a feedforward network with multiple hidden layers. The network has an input layer, an output layer and a few hidden layers. Each layer performs a distinct set of features in each layer. The complexity of the layers allow it to recognize, recombine and aggregate features. The DNN is capable of handling high dimensional and large datasets of different non-linear functions.

4. Results and Discussion

The results from the training and testing of all the models mentioned above are tabulated in Table 1. The hardware environment includes a laptop with Intel 2.50 GHz CPU and 32G Memory. The training and, fine-tuning of the models were performed in Keras platform in Python 3.5. The Tensor flow was used as the Backend system. It allows the use of different optimizer functions. A few optimizer functions were used to minimize or maximize the objective function. For example, the Stochastic Gradient Descent (SGD) Optimiser was applied to update the parameters such as learning rate decay, momentum and Nesterov Momentum. The N5 Loudness (S5 Sharpness) that exceeded 5% percentile of the time dependent Loudness (Sharpness) curve were used.

As seen in Table 1, the performance of Deep Neural Network is considerably better than the other models. SoftMax, random forest model and decision tree have a lower RMSE in SPL, Loudness and Sharpness. As observed in Table 1, the RMSE for Deep Neural Network is not more than 2dBA. Conversely, the classical Feedforward Neural Networks using backpropagation does not perform well. It is mainly due to the simple network structure with less nonlinear functions used. As seen in Figure 2, the Deep Neural Network gives the best results for the RMSE in both training and testing at around 100 epochs.

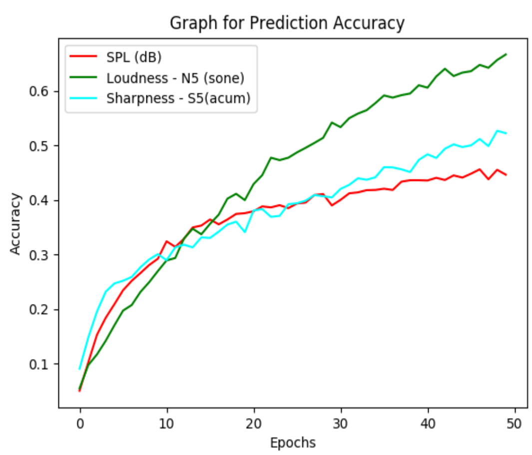
Figure 2 shows the graph for prediction accuracy over a number of epochs or iterations. The Keras module in the Python was used for visualizing the results. From the graph, it is evident that the accuracy increases over the number of iterations. However, the improvement in the accuracy is minimal once it reaches certain number of iterations. Sometimes, the accuracy decreases. Hence, the challenge is to optimize the model to obtain the optimal number for training iterations. At around 300 to 500 epochs for the Loudness and Sharpness (about 2500 to 5000 for SPL), the improvement in the accuracy is quite marginal. The maximum test accuracy of the prediction reached about 85% to 90%. In this paper, the training epoch was set to 300 for the Loudness and Sharpness and 2500 for the SPL. In summary, the RMSE of Deep Neural Network is approximately 2 dB(A). The RMSE of Loudness and Sharpness are around 2 sone and 3 acum, respectively.

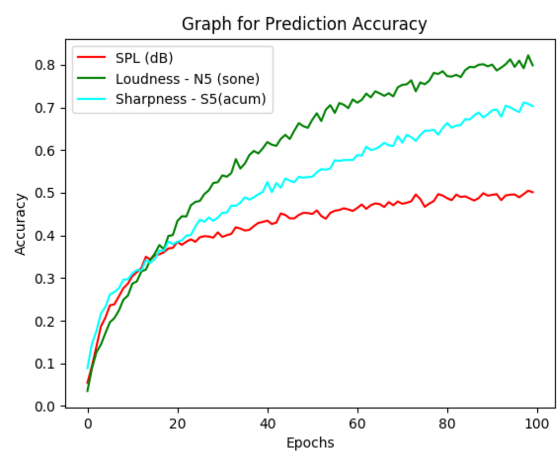
5. Conclusions

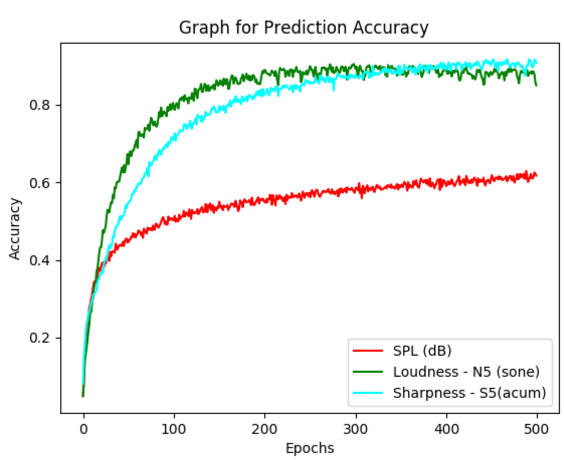
The prediction model using Artificial Neural Network (ANN) was developed to predict sound pressure level (SPL), Loudness and Sharpness for different room types. Several ANN models were compared and study to predict the SPL, Loudness, and Sharpness that were measured from different rooms. The measurements were performed with a class 1 certified and calibrated sound meter. The results from the comparison shown that the results from the Deep Neural Network had a higher accuracy and lower root mean square error. The final model using the Deep Neural Network can predict the SPL with less than 2 dB(A) root mean square error. On the other hand, the RMSE of Loudness and Sharpness were around 2 sone and 3 acum, respectively. Future works will include the use of other ANNs on more dataset.

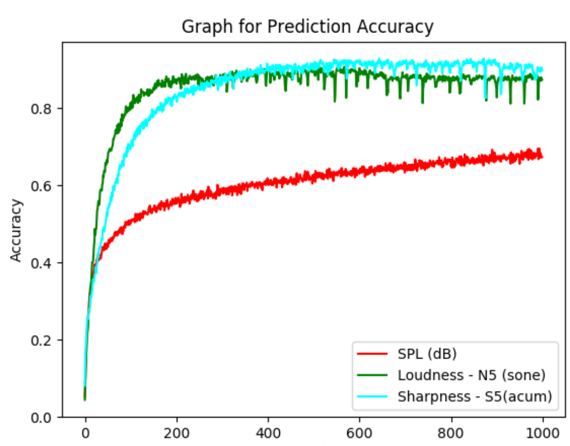
Table 1: Comparisons of different neural networks

			Testing root mean square error (RMSE)		
Models	Activation Functions	No of Neurons	Sound Pressure Level (dBA)	Loudness –N5(sone)	Sharpness – S5 (acum)
Feedforward Neural Networks with classical back-propagation	Sigmoid	100	56.44	28.06	57.31
Logistics Regression	--	--	19.77	17.31	10.83
K-Nearest Neighbours	--	100	19.96	19.82	11.84
Decision Tree	--	100	19.85	11.58	8.83
Support Vector Machine	--	--	14.13	14.13	14.11
Extreme Learning Machine	Sigmoid	100	19.78	17.31	11.07
Random Forest	--	100	19.91	4.24	7.94
Auto Encoders	ReLU	100	19.92	19	11.21
SoftMax	Softmax	100	8.64	13.74	8.76
Deep Neural Network	ReLU	100	**1.91**	**2.16**	**3.11**









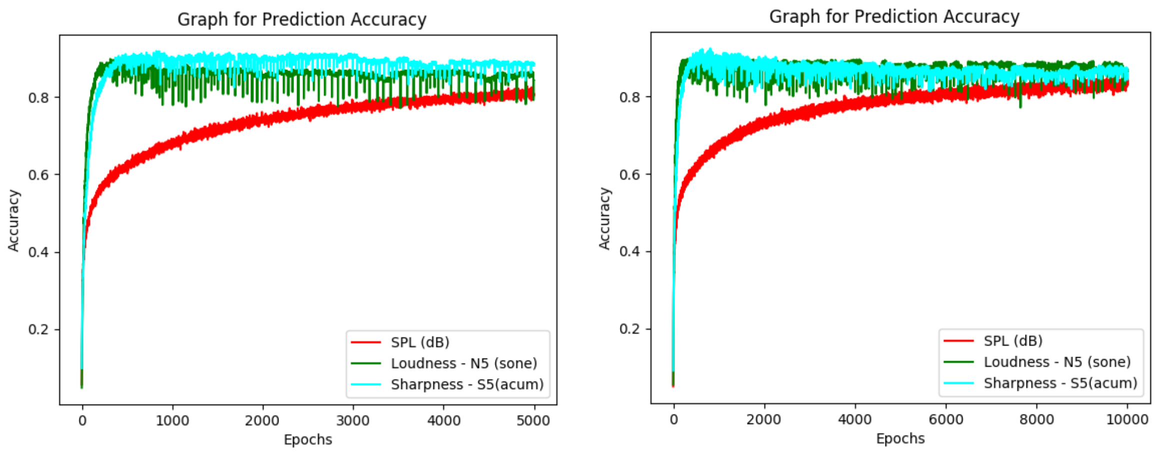


Figure 2: Prediction accuracy during training using different epochs for Deep Neural Network

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